



## AQ9.6: Activity Questions 6 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1

1 point

Consider a function  $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ .

Choose the correct statement with respect to  $\nabla f$ .

☐

$\nabla f$  has co-domain  $\mathbb{R}^2$

☐

$\nabla f$  has co-domain  $\mathbb{R}$

☐

$\nabla f$  has co-domain  $\mathbb{R}^3$

☐

$\nabla f$  has domain  $\mathbb{R}^3$

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

$\nabla f$  has co-domain  $\mathbb{R}$

$\nabla f$  has domain  $\mathbb{R}^3$

1 point

Let  $f: \mathbb{R}^3 \rightarrow \mathbb{R}$  and suppose its gradient  $\nabla f$  is continuous. Which of the following statements is true for the directional derivative of  $f$  in the direction of  $a \in \mathbb{R}^3$ ? Consider dot product to be the inner product  $\langle \cdot, \cdot \rangle$ .

☐

$D_a f = \frac{1}{\|a\|} \langle a, \nabla f \rangle$

☐

$D_a f = \langle a, \nabla f \rangle$

☐

$D_a f = \frac{1}{2\|a\|} \langle a, \nabla f \rangle$

☐

$D_a f = \frac{1}{\|a\|^2} \langle a, \nabla f \rangle$

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

$D_a f = \frac{1}{\|a\|} \langle a, \nabla f \rangle$

Let  $f(x, y) = e^x \sin y$ . If  $\nabla f$  at the point  $(\ln(2), \pi/4)$  is  $(k, k)$  where  $k \in \mathbb{R}$ , then find the value of  $k^2$ .

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

(Type: Numeric) 2

1 point

### Level 2



1 point

Let  $p = -\frac{k}{(\|r\|)^3}r$ , where  $r = (x, y, z) \in \mathbb{R}^3 \setminus \{(0, 0, 0)\}$ ,  $k \in \mathbb{R}$ .  
 $(x, y, z) \in \mathbb{R}^3 \setminus \{(0, 0, 0)\}$ ,  $k \in \mathbb{R}$ .

Amongst the options below, choose those for which the function  $f$  satisfies  $\nabla f = p \nabla f = p$  for some  $k \in \mathbb{R}$ .

Consider dot product as the inner product for finding the norm.



$$f(x, y, z) = \frac{1}{\|r\|} f(x, y, z) = \|r\|$$



$$f(x, y, z) = \frac{1}{2\|r\|} f(x, y, z) = 2\|r\|$$



$$f(x, y, z) = \frac{1}{\|r\| + 2} f(x, y, z) = \|r\| + 2$$



$$f(x, y, z) = \|r\| f(x, y, z) = \|r\|$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$f(x, y, z) = \frac{1}{\|r\|} f(x, y, z) = \|r\|$$

$$f(x, y, z) = \frac{1}{2\|r\|} f(x, y, z) = 2\|r\|$$

1 point

Let  $f(x, y, z) = 4(x^2 + y^2) - z^2$ ,  $f(x, y, z) = 4(x^2 + y^2) - z^2$ .

Choose the correct option(s).



$$\nabla f \nabla f \text{ at } (1, 0, 2) \text{ is } (8, 0, -4)$$



$\nabla f \nabla f$  is continuous everywhere in  $\mathbb{R}^3$



$\nabla f \nabla f$  is not continuous everywhere in  $\mathbb{R}^3$



$$\nabla f \nabla f \text{ at } (1, 0, 2) \text{ is } (-8, 0, -4)$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\nabla f \nabla f \text{ at } (1, 0, 2) \text{ is } (8, 0, -4)$$

$\nabla f \nabla f$  is continuous everywhere in  $\mathbb{R}^3$

1 point

Consider a function defined as

$$f(x, y) = \begin{cases} \frac{y^3}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases} \quad f(x, y) = \begin{cases} x^2 + y^2 y^3 & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Choose the correct option(s).



$$\frac{\partial f}{\partial x} = \frac{-2xy^3}{(x^2 + y^2)^2}, \quad \frac{\partial f}{\partial x} = (x^2 + y^2)^2 - 2xy^3, \text{ for } (x, y) \neq (0, 0)$$

☐

$$\frac{\partial f}{\partial x} = \frac{2xy^3}{(x^2 + y^2)^2}, \partial_x \partial f = (x^2 + y^2)^{-2} 2xy^3, \text{ for } (x, y) \neq (0, 0) \quad (x, y) = (0, 0)$$

☐

$$\frac{\partial f}{\partial y} = \frac{3x^2y^2 + y^4}{(x^2 + y^2)^2} \partial_y \partial f = (x^2 + y^2)^{-2} 3x^2y^2 + y^4 \text{ for } (x, y) \neq (0, 0) \quad (x, y) = (0, 0)$$

☐

$$\frac{\partial f}{\partial y} = \frac{3x^2y^2 - y^4}{(x^2 + y^2)^2} \partial_y \partial f = (x^2 + y^2)^{-2} 3x^2y^2 - y^4 \text{ for } (x, y) \neq (0, 0) \quad (x, y) = (0, 0)$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\frac{\partial f}{\partial x} = \frac{-2xy^3}{(x^2 + y^2)^2}, \partial_x \partial f = (x^2 + y^2)^{-2} - 2xy^3, \text{ for } (x, y) \neq (0, 0) \quad (x, y) = (0, 0)$$

$$\frac{\partial f}{\partial y} = \frac{3x^2y^2 + y^4}{(x^2 + y^2)^2} \partial_y \partial f = (x^2 + y^2)^{-2} 3x^2y^2 + y^4 \text{ for } (x, y) \neq (0, 0) \quad (x, y) = (0, 0)$$

1 point

Consider the function definition given in Q6 above. Choose the correct option(s).

☐

$$\frac{\partial f}{\partial x} \partial_x \partial f \text{ is continuous at } (0, 0) \quad (0, 0)$$

☐

$$\frac{\partial f}{\partial x} \partial_x \partial f \text{ is not continuous at } (0, 0) \quad (0, 0)$$

☐

$$\frac{\partial f}{\partial y} \partial_y \partial f \text{ is not continuous at } (0, 0) \quad (0, 0)$$

☐

$$\nabla f \nabla f \text{ is discontinuous at the origin } (0, 0) \quad (0, 0)$$

☐

Directional derivative of  $f$  in the direction of a vector  $a \in \mathbb{R}^2$  at  $(0, 0)$  can be obtained using the gradient of  $f$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\frac{\partial f}{\partial x} \partial_x \partial f \text{ is not continuous at } (0, 0) \quad (0, 0)$$

$$\frac{\partial f}{\partial y} \partial_y \partial f \text{ is not continuous at } (0, 0) \quad (0, 0)$$

$$\nabla f \nabla f \text{ is discontinuous at the origin } (0, 0) \quad (0, 0)$$

1 point

Consider a function defined as

$$f(x, y) = \begin{cases} (x^2 + y^2) \sin\left(\frac{1}{\sqrt{x^2 + y^2}}\right) & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases} \quad f(x, y) = \begin{cases} (x^2 + y^2) \sin\left(\frac{1}{\sqrt{x^2 + y^2}}\right) & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$



$$f(0, 0) = 0 \quad (x, y) = (0, 0).$$

Choose the correct option(s).

☐

$$\frac{\partial f}{\partial y}(0, 0) = 0 \quad \partial_y \partial f(0, 0) = 0$$

☐

$$\frac{\partial f}{\partial x}(0, 0) = 0 \quad \partial_x \partial f(0, 0) = 0$$

☐

$$\frac{\partial f}{\partial x} = 0 \quad \partial_x \partial f = 0$$

☐

$$\frac{\partial f}{\partial y} = 0 \Rightarrow \frac{\partial}{\partial y} \frac{\partial f}{\partial x} = 0$$

☐ None of the above

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\frac{\partial f}{\partial y}(0, 0) = 0 \Rightarrow \frac{\partial}{\partial y} \frac{\partial f}{\partial x}(0, 0) = 0$$

$$\frac{\partial f}{\partial x}(0, 0) = 0 \Rightarrow \frac{\partial}{\partial x} \frac{\partial f}{\partial y}(0, 0) = 0$$

1 point

Consider the function definition given in Q8 above. Choose the correct option(s) with respect to the partial derivative of  $f$ . Note  $\text{sign}(x) = \pm 1$  depending on the sign of  $x$ .

☐

$$\frac{\partial f}{\partial x}(x, 0) = 2x \sin\left(\frac{1}{|x|}\right) - \text{sign}(x) \cos\left(\frac{1}{|x|}\right) \Rightarrow \frac{\partial}{\partial x} \frac{\partial f}{\partial y}(x, 0) = 2x \sin(|x|) - \text{sign}(x) \cos(|x|) \text{ for } x \neq 0$$

☐

$$\frac{\partial f}{\partial x}(x, 0) = 2x \sin\left(\frac{1}{|x|}\right) - \text{sign}(x) \cos\left(\frac{1}{|x|}\right) \Rightarrow \frac{\partial}{\partial x} \frac{\partial f}{\partial y}(x, 0) = 2x \sin(|x|) - \text{sign}(x) \cos(|x|)$$

☐

$$\frac{\partial f}{\partial x}(x, 0) = 2x \sin\left(\frac{1}{|x|}\right) - \cos\left(\frac{1}{|x|}\right) \Rightarrow \frac{\partial}{\partial x} \frac{\partial f}{\partial y}(x, 0) = 2x \sin(|x|) - \cos(|x|) \text{ for } x \neq 0$$

☐

$$\frac{\partial f}{\partial y}(0, y) = 2y \sin\left(\frac{1}{|y|}\right) - \text{sign}(y) \cos\left(\frac{1}{|y|}\right) \Rightarrow \frac{\partial}{\partial y} \frac{\partial f}{\partial x}(0, y) = 2y \sin(|y|) - \text{sign}(y) \cos(|y|) \text{ for } y \neq 0$$

☐

$$\frac{\partial f}{\partial y}(0, y) = 2y \sin\left(\frac{1}{|y|}\right) - \text{sign}(y) \cos\left(\frac{1}{|y|}\right) \Rightarrow \frac{\partial}{\partial y} \frac{\partial f}{\partial x}(0, y) = 2y \sin(|y|) - \text{sign}(y) \cos(|y|)$$

☐

$$\frac{\partial f}{\partial y}(0, y) = 2y \sin\left(\frac{1}{|y|}\right) - \cos\left(\frac{1}{|y|}\right) \Rightarrow \frac{\partial}{\partial y} \frac{\partial f}{\partial x}(0, y) = 2y \sin(|y|) - \cos(|y|) \text{ for } y \neq 0$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\frac{\partial f}{\partial x}(x, 0) = 2x \sin\left(\frac{1}{|x|}\right) - \text{sign}(x) \cos\left(\frac{1}{|x|}\right) \Rightarrow \frac{\partial}{\partial x} \frac{\partial f}{\partial y}(x, 0) = 2x \sin(|x|) - \text{sign}(x) \cos(|x|) \text{ for } x \neq 0$$

☐

$$\frac{\partial f}{\partial y}(0, y) = 2y \sin\left(\frac{1}{|y|}\right) - \text{sign}(y) \cos\left(\frac{1}{|y|}\right) \Rightarrow \frac{\partial}{\partial y} \frac{\partial f}{\partial x}(0, y) = 2y \sin(|y|) - \text{sign}(y) \cos(|y|) \text{ for } y \neq 0$$

☐

1 point

Consider the function definition given in Q8 above. Choose the correct option.

☐

$\nabla f$  is continuous everywhere

☐

$\nabla f$  is discontinuous at the origin  $(0, 0)$

☐

$\frac{\partial f}{\partial x}$  is continuous at the origin but  $\frac{\partial f}{\partial y}$  is discontinuous at the origin

☐

$\frac{\partial f}{\partial y}$  is continuous at the origin but  $\frac{\partial f}{\partial x}$  is discontinuous at the origin

☐

Directional derivative of  $f$  in the direction of a vector  $a \in \mathbb{R}^2$  at  $(0, 0)$  cannot be obtained using the gradient of  $f$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\nabla f$  is discontinuous at the origin  $(0, 0)$

Directional derivative of  $f$  in the direction of a vector  $a \in \mathbb{R}^2$  at  $(0, 0)$  cannot be obtained using the gradient of  $f$

Your score is: 0/10